Lab 5: Minimax & Alpha-Beta Pruning — Submission

Code: `lab5_minimax_alphabeta.py` (run with `python3 lab5_minimax_alphabeta.py`)

Task 1: Hand trace (Minimax)

Tree 5.1 — scores `[3, 5, 2, 9, 0, 7]`, padded with zeros to 8 leaves `[3, 5, 2, 9, 0, 7, 0, 0]`, depth 3 (root = MAX, then MIN, MAX, leaves).

Level	Type	Nodes (left → right)	Computation	Values
3	leaves			3, 5, 2, 9, 0, 7, 0, 0
2	MAX	A, B, C, D	max(3,5), max(2,9), max(0,7), max(0,0)	5, 9, 7, 0
1	MIN	E, F	min(5,9), min(7,0)	5, 0
0	MAX	root	max(5,0)	**5**

Program output: `The optimal value is: 5` (matches). Best action: the left branch. Note the padded zeros are artificial leaves, so they influence node D and F; with real game data you would pad with a neutral value for the right player or use an unbalanced tree.

Tree 5.2 — `[3, 5, 2, 9, 3, 5, 2, 9]`: MAX level gives 5, 9, 5, 9; MIN level gives 5, 5; root = **5**.

Task 2: Leaf-evaluation counters

A one-element list `counter` is incremented in the terminal case of both functions.

Tree	Root value	Minimax leaves	Alpha-beta leaves
5.1 (padded)	5	8	8
5.2	5	8	6

Alpha-beta trace on 5.2 (α, β shown when entering each node):

Step	Node	(α, β)	Action
1	MIN-left	(−∞, +∞)	explore
2	MAX A	(−∞, +∞)	leaves 3, 5 → 5; MIN-left β = 5
3	MAX B	(−∞, 5)	leaf 2 → 2; leaf 9 → 9; 9 ≥ β (cut-off, nothing left to skip) → MIN-left = 5; root α = 5
4	MIN-right	(5, +∞)	explore
5	MAX C	(5, +∞)	leaves 3, 5 → 5; MIN-right β = 5

6	MAX D	(5, 5)	**pruned**: α ≥ β, leaves 2 and 9 never evaluated
7	root		max(5, 5) = **5**

Two leaves skipped (6 of 8 evaluated). On 5.1 nothing prunes: the right MIN node's β only drops to 7 while α = 5, so α < β throughout.

Task 3: Reordering branches

Reordering means swapping the children of a node (the tree's structure and leaf set stay the same, so the Minimax value cannot change).

Reordering of 5.2	Leaves	Root	Alpha-beta leaves
Original	3,5,2,9,3,5,2,9	5	6
Swap the 4 leaf pairs	5,3,9,2,5,3,9,2	5	**5**
Swap children of both MIN nodes	2,9,3,5,2,9,3,5	5	**8** (no pruning)
Swap children of the root	same leaves	5	6
Swap children at every node	9,2,5,3,9,2,5,3	5	8

Exhaustive check: all $2^7 = 128$ child-swap orderings of the tree were run. Root value was 5 in every one; alpha-beta leaf counts were 5 (16 orderings), 6 (48), 7 (16), 8 (48). Alpha-beta never evaluated more leaves than plain Minimax (always 8).

Edge case / limitation

- **All-equal leaves** (`[4]*8`): value 4, alpha-beta evaluates 5 leaves. The `>=` test prunes even on ties, which is safe because the value cannot improve.
- Pruning depends entirely on ordering. Worst-case ordering gives no savings (8/8); good ordering (best moves first) gives the biggest cut. Real engines use heuristics (killer moves, iterative deepening) for this.
- The code assumes a complete binary tree of fixed depth.

Interpretation (link to learning outcome)

Alpha-beta returns the same root value as Minimax in every test, confirming that pruning only skips branches that cannot influence the decision. α is the best value MAX can already guarantee, β the best MIN can guarantee; once α ≥ β the current node can't affect the result because the opponent above would never allow it.

Viva-ready design choices

- **Counter in the base case**: counts each terminal evaluation exactly once, so comparison is fair.
- **`>=` in the cut-off**: pruning on equality is safe for the root value (not for finding *all* optimal moves).
- **Sibling swaps for Task 3**: swapping arbitrary leaves between subtrees changes the game itself, so the value would change. Only sibling reordering preserves it.

Reflection

- **Learned:** pruning efficiency comes from ordering, not from the algorithm alone — identical trees range from 5 to 8 evaluations.
- **Real use:** game engines, planning against adversaries; combine with move ordering and depth limits plus an evaluation function.
- **Surprise:** my first attempt at "reordering" shuffled leaves across subtrees, which changed the root value (3, 5 or 9). That showed why only sibling swaps count as reordering.
- **Next time:** test unbalanced trees and add the chosen action (not just the value) to the output.